

Problem 1 : Subadditive Sequences

Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive real numbers such that

$$a_{n+m} \leq a_n + a_m, \quad m, n \geq 1.$$

Show that $\lim_{n \rightarrow \infty} \frac{a_n}{n}$ exists by showing

$$\lim_{n \rightarrow \infty} \frac{a_n}{n} = \inf_{n \geq 1} \frac{a_n}{n}.$$

Hint. Treat separately $\liminf$ and $\limsup$.

Solution.

Problem 2 : Alternating Series of Continuous Functions

Let $f_n : [-1, 1] \rightarrow [0, 1]$ be continuous functions. Assume that for each $x \in [-1, 1]$, the sequence $(f_n(x))_{n \geq 1}$ is decreasing and

$$\lim_{n \rightarrow \infty} f_n(x) = 0.$$

For $n \geq 1$ and $x \in [-1, 1]$, let

$$g_n(x) = \sum_{m=1}^n (-1)^m f_m(x).$$

Show that $g_n(x)$ converges to some $g(x) \in \mathbb{R}$ for all $x \in [-1, 1]$, and that the function $g : [-1, 1] \rightarrow \mathbb{R}$ thus defined is continuous on $[-1, 1]$.

Solution.

By Dini's Theorem: $f_n \rightarrow 0$ uniformly.

By the alternating series test. $g_n(x) \rightarrow g(x)$ pointwise for all $[-1, 1]$. $\forall \varepsilon > 0$, there exists a $N \in \mathbb{N}$ such that $\forall n \geq N$: $\|f_n\|_{\infty} < \frac{\varepsilon}{3}$ for all $x \in [-1, 1]$. By the alternating series estimation theorem:

$$|g_n(x) - g(x)| \leq |g_{n+1}(x) - g_n(x)| = f_{n+1}(x) < \varepsilon.$$

Since f_N is continuous, $\exists \delta > 0$ such that $|f_N(x) - f_N(y)| < \frac{\varepsilon}{3}$ if $|x - y| < \delta$. $\forall x, y \in [-1, 1]$ if $d(x, y) < \delta$:

$$|g(x) - g(y)| \leq |g(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - g(y)| < \varepsilon.$$

Thus g is continuous.

Problem 3

Let (X, d_X) be a compact metric space.

(a) Show that

$$d_Y(f, g) = \sum_{n \in \mathbb{Z}} 2^{-|n|} d_X(f(n), g(n))$$

defines a metric on $Y = \{f : \mathbb{Z} \rightarrow X\}$.

(b) Show that (Y, d_Y) is compact.

Solution.

(a): $\forall f \in Y: \forall n \in \mathbb{Z}, d_Y(f(n), f(n)) = 0$ thus

$$d_Y(f, f) = \sum_{n \in \mathbb{Z}} 2^{-|n|} d_X(f(n), g(n)) = 0.$$

Naturally positive and symmetric.

$\forall f, g, h \in Y:$

$$\begin{aligned} d_Y(f, g) &= \sum_{n \in \mathbb{Z}} 2^{-|n|} d_X(f(n), g(n)) \\ &\leq \sum_{n \in \mathbb{Z}} 2^{-|n|} [d_X(f(n), h(n)) + d_X(h(n), g(n))] = d_Y(f, h) + d_Y(h, g) \end{aligned}$$

(b): Let $(f_n)_{n=0}^{\infty}$ be a sequence in Y . Let $\mathbb{Z} = \{z_1, z_2, \dots\}$ be an enumeration of $\mathbb{Z}$. We will do a diagonalization argument to get a subsequence.

Let $g_n^0 = f_n$. $(g_n^0(z_1))_{n=1}^{\infty}$ is a sequence in X , and by compactness of X , there exists a subsequence $(g_n^1)_{n \in \mathbb{N}}$ such that $(g_n^1(z_1))_{n \in \mathbb{N}}$ is convergent. Inductively create g_n^i such that g_n^i is a subsequence of g_n^{i-1} and converges pointwise on $\{z_1, \dots, z_i\}$.

Let $h_n = g_n^n$. The tail of h_n is a subsequence of g_n^n and hence converges on $\{z_1, \dots, z_n\}$. Hence h_n has pointwise convergence on $\mathbb{Z}$.

$\forall \varepsilon > 0$: There is a $\frac{\varepsilon}{3}$ -net on X .

$$d(h_n(n), h(n)) \leq d(h_n(n), c_i) + d(c_i, h(n))$$

Problem 4 : Increasing Functions and Uniform Convergence

Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be increasing continuous functions. Assume that f_n converge pointwise to a continuous function $f : [0, 1] \rightarrow \mathbb{R}$. Prove that f_n converge uniformly to f .

Solution. Since $[0, 1]$ is compact, f_n and f are uniformly continuous. $\forall \varepsilon > 0$, there exists a $\delta > 0$ such that $\forall x, y \in [0, 1]$ where $0 < x - y < \delta$ implies that $0 < f(x) - f(y) < \varepsilon$. By total boundedness of $[0, 1]$, There exists $0 \leq a_1 < a_2 < \dots < a_n \leq 1$ such that

$$[0, 1] = \bigcup_{i=1}^n B_{\frac{\delta}{2}}(a_i).$$

Note $a_{i-1} - a_i < \delta$. $\forall i \in \{1, \dots, n\}$, $\exists N_i \in \mathbb{N}$ such that for all $m \geq N$: $|f_m(a_i) - f(a_i)| < \varepsilon$. Let $N = \max\{N_i : 0 \leq i \leq n\}$. $\forall x \in [0, 1]$ there exists a $i \in \{1, \dots, n\}$ such that $x \in [a_i, a_{i+1}]$. f is a increasing continous function. By monotonicity, we have that $\forall m \in \mathbb{N}$,

$$f_m(a_i) \leq f_m(x) \leq f_m(a_{i+1})$$

For $m \geq N$: $f_m(a_i) - f(a_i) > -\varepsilon \Rightarrow f(a_i) - \varepsilon < f_m(a_i)$. Similarly we see $f_m(a_{i+1}) < f(a_{i+1}) + \varepsilon$. Thus

$$f(a_i) - \varepsilon \leq f_m(x) \leq f(a_{i+1}) + \varepsilon.$$

Thus

$$\forall m \geq N$$

:

$$\begin{aligned} |f_m(x) - f(x)| &= |f_m(x) - f(a_i) + f(a_i) - f(a_{i+1}) + f(a_{i+1}) - f(x)| \\ &\leq |f_m(x) - f(a_i)| + |f(a_{i+1}) - f(a_i)| + |f(a_i) - f(x)| \\ &< \varepsilon + \varepsilon + \varepsilon = 3\varepsilon \end{aligned}$$

Problem 5 : Hölder Continuous Functions

Let $f : [0, 1] \rightarrow \mathbb{R}$. We say that f is Hölder continuous of order $\alpha \in (0, 1)$ and write $f \in C^\alpha([0, 1])$ if

$$\|f\|_{C^\alpha} := \sup\{|f(x)|; x \in [0, 1]\} + \sup\left\{\frac{|f(x) - f(y)|}{|x - y|^\alpha}; x, y \in [0, 1], x \neq y\right\} < \infty.$$

For $f, g \in C^\alpha([0, 1])$, define $d(f, g) = \|f - g\|_{C^\alpha}$.

(a) Show that $(C^\alpha([0, 1]), d)$ is a complete metric space.

(b) Let (f_n) be a sequence in $C^{1/2}([0, 1])$ such that $\|f_n\|_{C^{1/2}} \leq 1$ for $n = 1, 2, \dots$. Show that (f_n) has a subsequence that converges in $C^{1/3}([0, 1])$.

Solution.